Unrotating palaeocurrent measurements (Barrême data)

The following are a number of palaeocurrent measurements taken from the bases of beds in the Bussière d'Entouart (BdE), Laubre (Lau), and L'Asse (Asse) Formations/Members, near Clumanc in the Barrême basin.

locality	*unit*	*bedding*	*type*	*pl/pl dir*	*direction*	*unrotated*
22	BdE	003/29E	scours	10/162	N	342°
49	BdE	154/44SW	scours	11/166	N	347°
114	BdE	007/25E	scours	9/026	N	030°
114	BdE	021/17E	scours	6/177	N	358°
114	BdE	005/27E	scours	11/161	N	340°
116	BdE	026/15E	scours	4/187	N	008°
138 (N)	Lau	179/37E	furrows	35/112	E	111°
138 (N)	Lau	179/37E	furrows	34/118	E	116°
138 (N)	Lau	179/37E	furrows	34/115	E	113°
138 (N)	Lau	179/37E	furrows	31/127	E	124°
27	Lau	176/43E	furrows	18/155	SE	152°
27 (N)	Lau	179/35E	scours	18/153	SE	150°
27 (N)	Lau	179/35E	scours	15/158	SE	156°
11-12	Asse	008/24E	scours	23/113	W	294°
11-12	Asse	008/24E	scours	23/116	W	296°
11-12	Asse	008/24E	scours	23/111	W	292°
12	Asse	006/26E	scours	24/125	NW	304°
12	Asse	006/26E	scours	26/105	W	285°
12	Asse	006/26E	scours	23/130	NW	310°
12	Asse	006/26E	scours	19/145	NW	324°
12	Asse	006/26E	scours	25/109	W	290°
38	Asse	015/20E	scours	17/140	SE	140°
136-137	Asse	012/22E	scours	12/043	SW	227°
136-137	Asse	012/22E	scours	19/069	W	253°
139 (N)	Asse	014/21E	scours	20/081	E	084°

(a) Plot a stereogram of all the bedding readings and determine the plunge/plunge direction of the fold that is implied by this dataset.

(b) Determine the original palaeocurrent directions before folding for the L'Asse data. The original palaeocurrent direction before folding has been determined for you for the data in the other two units

(c) Plot a rose diagram showing the distribution of original palaeocurrent directions in the L'Asse Member

(d) Add the data from the other two units to your rose diagram using a different colour for each unit.

A handout explaining how to do a statistical analysis of directional data (working out mean directions and variances) is on Canvas